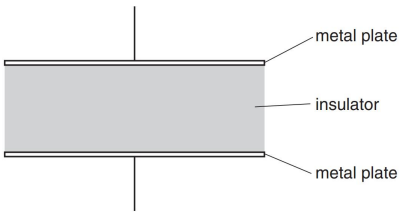
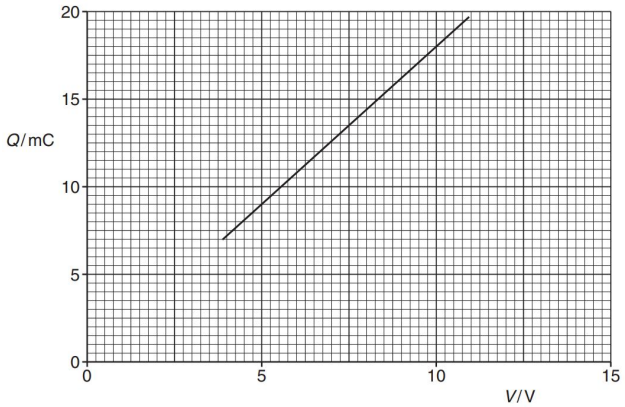
19.1 Exercises - Capacitors and capacitance

1. A capacitor consists of two metal plates separated by an insulator, as shown in Fig.





The potential difference between the plates is V. The variation with V of the magnitude of the charge Q on one plate is shown in Fig.

a. Define the capacitance of the capacitor.

.....

..... [2]

b. Use Fig. to determine

i. the capacitance of the capacitor,

capacitance = ...μF [2]

ii. he loss in energy stored in the capacitor when the potential difference V is reduced from 10.0 V to 7.5 V.

energy = ... mJ [2]

c. Explain why the capacitor stores energy but not charge.

.....

.....

..... [3]

d. The capacitor is charged so that the potential difference between its plates is V_0.

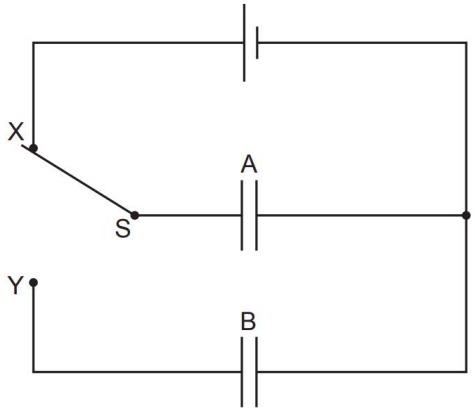
The capacitor is then connected across a resistor for a short time. It is then disconnected.

The energy stored in the capacitor is reduced to $\frac{1}{16}$ of its initial value.

Determine, in terms of V_0, the potential difference across the capacitor.

potential energy = ... [2]

2. Two capacitors A and B are connected into the circuit shown in Fig.



Capacitor A has capacitance C and capacitor B has capacitance 3C.

The electromotive force (e.m.f.) of the cell is V.

The two-way switch S is initially at position X, and capacitor B is initially uncharged.

a. State, in terms of V and C, expressions for:

i. the initial charge Q_A on the plates of capacitor A

Q_A = ... [1]

ii. the initial energy E_A stored in capacitor A.

E_A = ... [1]

b. The two-way switch S is now moved to position Y.

i. State and explain what happens to the charge that was initially on the plates of capacitor A.

.....

.....[2]

ii. What is the relation between electromotive force (e.m.f.) of V and the final potential difference (p.d.) V_B

..... [3]

iii. Determine an expression, in terms of V and C, for the decrease ΔE in the total energy that is stored in the capacitors as a result of the change of the position of the switch.

ΔE = ... [2]